ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** O	O.
2	**(d)** iron	Iron.
3	**(b)** wires	Wire drawing.
4	**(d)** H	H.
5	**(a)** sulphur	Sulphur.
6	**(c)** sheets	Sheets.
7	**(c)** C	C.
8	**(c)** mercury	Mercury.
9	**(d)** lustre	Shine.
10	**(a)** Ca	Ca.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pure substance made of one kind of atom (2).	2
12	Iron and copper (1); Sulphur and carbon (1).	1+1
13	Brass (1); Steel (1).	1+1
14	Substance of two or more elements chemically combined (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	H, O (1); C, N (1); S, Ca (1).	1+1+1
16	Water (1); Salt (1); Carbon dioxide (1).	1+1+1
17	Fixed composition (1); New properties (1); Separated by chemical methods (1).	1+1+1
	OR: Iron (1); Copper (1); Zinc (1).	1+1+1
18	Hydrogen (1); Oxygen (1); Nitrogen (1).	1+1+1
	OR: No fixed composition (1); Properties of components stay (1); Separated by physical methods (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Composition (1); Types of atoms (1); Breaking down (1); Examples (1); Symbols/formulae (1).	1+1+1+1+1
	OR: Uniform composition (1); Non-uniform (1); Salt water, air, brass (1); Sand-water, oil-water, soil (1); Different separation methods (1).	1+1+1+1+1
20	Composition (1); Separation (1); Properties (1); Energy change (1); Examples (1).	1+1+1+1+1
	OR: Nitrogen about 78 per cent (1); Oxygen about 21 per cent (1); Argon, CO2 (1); Water vapour (1); Physical separation possible (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a molecule such as O_2 a compound; a compound needs *different* elements.
- Saying air or an alloy is a compound; they are mixtures.
- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.
- Writing that all non-metals are poor conductors; graphite is an exception.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** element	Element.
2	**(b)** copper	Metal.
3	**(b)** Cl	Cl.
4	**(c)** compound	Compound.
5	**(d)** sulphur	Non-metal.
6	**(b)** Cu	From cuprum.
7	**(d)** mixture	Many gases.
8	**(c)** copper and zinc	Alloy.
9	**(a)** Au	From aurum.
10	**(b)** mixture	Water and salts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Shining surface of metals (2).	2
12	Salt solution (1); Air (1).	1+1
13	Two or more substances physically mixed (2).	2
14	Mercury (1); Bromine (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lustre (1); Malleable (1); Conduct electricity (1).	1+1+1
16	Iron for bridges (1); Copper for wires (1); Aluminium for foil (1).	1+1+1
17	Atom: smallest particle of an element, e.g. H (1); Molecule: group of atoms, e.g. O2 (1); Molecules of compounds like H2O (1).	1+1+1
	OR: Dull (1); Brittle (1); Poor conductors (1).	1+1+1
18	Oxygen for breathing (1); Carbon as fuel (1); Sulphur in medicines (1).	1+1+1
	OR: H2O: 3 (1); CO2: 3 (1); O2: 2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lustre (1); Malleability (1); Ductility (1); Conductivity (1); Sonority (1).	1+1+1+1+1
	OR: Magnet removes iron (1); Add water (1); Filter to remove sand (1); Evaporate for salt (1); Name physical methods (1).	1+1+1+1+1
20	Dull (1); Brittle (1); Poor conductors (1); Low density (1); Exceptions: graphite, bromine (1).	1+1+1+1+1
	OR: Water H2O (1); Carbon dioxide CO2 (1); Salt NaCl (1); Ammonia NH3 (1); Methane CH4 (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** carbon	Alloy.
2	**(d)** Ag	From argentum.
3	**(c)** homogeneous	Uniform.
4	**(d)** hydrogen and oxygen	H and O.
5	**(a)** H_2O	Water.
6	**(c)** heterogeneous mixture	Not uniform.
7	**(d)** sodium and chlorine	NaCl.
8	**(d)** CO_2	CO2.
9	**(b)** magnet	Magnetic.
10	**(a)** metalloid	Semi-conductor.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sand and water (1); Oil and water (1).	1+1
12	Fe (1); Na (1).	1+1
13	Mixture of metals or metal and non-metal (2).	2
14	Sodium chloride (1); Common salt (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Element: one kind of atom (1); Compound: two or more elements (1); Compounds can be broken down chemically (1).	1+1+1
16	Bring a magnet (1); Iron sticks (1); Sand remains (1).	1+1+1
17	Brass: copper, zinc (1); Steel: iron, carbon (1); Bronze: copper, tin (1).	1+1+1
	OR: Air (1); Sea water (1); Soil (1).	1+1+1
18	Heat to evaporate water (1); Salt remains (1); Collect the salt (1).	1+1+1
	OR: Wear goggles (1); Do not taste (1); Wash hands after use (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mixture heated (1); Compound formed (1); Black solid (1); No attraction to magnet (1); New properties (1).	1+1+1+1+1
	OR: H, He, C, N, O (2); Na, Mg, Al, S, Cl (2); All correct symbols (1).	2+2+1
20	Mixtures of metals (1); Brass (1); Steel (1); Bronze (1); Stronger and resistant (1).	1+1+1+1+1
	OR: Solids (1); Liquids (1); Gases (1); Examples (1); Symbols (1).	1+1+1+1+1

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